Predicting Golden Cross Transitions in ETFs: What Survives a Careful Evaluation

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Abstract

We study whether machine-learning classifiers can anticipate, up to 30 trading days ahead, crossovers between the 50-day and 200-day simple moving averages (Golden and Death Crosses) of exchange-traded funds. XGBoost and Random Forest classifiers are trained on SPY with 12 technical and moving-average features over 2000–2024 and applied without retraining to DIA, QQQ, GLD and TLT. Under a first evaluation protocol, in which the decision threshold is selected on the test set and confidence intervals assume independent observations, XGBoost reaches a mean F1 of 0.738 on SPY, DIA and QQQ. We then correct three weaknesses of this protocol. Selecting the threshold on a validation block inside the training period, with a 30-day embargo, lowers the mean F1 to 0.595 for XGBoost and 0.633 for Random Forest; almost all of this drop comes from the source of the threshold, and the embargo changes mean F1 by less than 0.02. A moving-block bootstrap that respects the 30-day autocorrelation of the labels makes the confidence intervals 2.6 to 3.7 times wider. Finally, a rule with no parameter fitted to data, which linearly extrapolates the gap between the two moving averages, obtains a mean F1 of 0.720 and the highest F1 on three of the five assets. A paired block bootstrap finds the rule better than both models on QQQ and better than XGBoost on TLT, and finds no asset where a model is significantly better than the rule. Pooled over the five assets, the models produce 3.0 (Random Forest) and 4.0 (XGBoost) times as many false-alarm days as the rule. Each test set contains only 5 to 10 crossover events, and XGBoost results move by up to 0.047 in mean F1 when prices are perturbed by one part in a million. We conclude that crossover timing over a 30-day horizon is largely predictable from the mechanics of the moving averages themselves, and that the apparent contribution of the classifiers is mostly an artefact of the first protocol.

1 Introduction

Moving-average crossover rules are among the most widely used trend indicators. They are usually studied as reactive signals: a position changes once the crossover has occurred (Avramov et al., 2018). The 50/200-day pair, known as the Golden Cross (upward) and Death Cross (downward), is the most common long-horizon variant, and the ratio between these two averages carries information about future returns (Bhootra, 2024). A natural question is whether the crossover itself can be anticipated.

This paper started as an attempt to answer that question with gradient boosting and random forests. A first round of experiments suggested a positive answer, with F1 scores around 0.74. Closer inspection showed that this figure depended on choices that a careful reviewer would reject. The paper therefore reports the full path from the first estimate to a corrected one.

Our contributions are:

1. a decomposition of how the evaluation protocol inflates performance, separating threshold selection on the test set from the train/test embargo, and a comparison of observation-level and block bootstrap intervals;

2. an extrapolation baseline with no parameter fitted to data that matches or exceeds the learned models, compared with them through a paired block bootstrap and event-level metrics, showing that the target is largely predictable by construction;

3. an account of the statistical limits of the setting: few independent events per test set, threshold instability across time, and sensitivity to negligible numerical perturbations.

Section 2 reviews related work. Section 3 describes the data and target. Section 4 presents the features, models and evaluation protocols. Section 5 reports results and Section 6 discusses limitations.

2 Related Work

The distance or ratio between short- and long-term moving averages captures changes in price regimes and predicts future returns (Avramov et al., 2018; Bhootra, 2024). Gradient boosting (Chen and Guestrin, 2016) and other tree ensembles perform competitively in stock direction prediction (Ampomah et al., 2020). Transitions between bull and bear phases tend to coincide with elevated volatility and trading activity (Catão and Timmermann, 2010), which motivates part of our feature set.

On the evaluation side, labels built from future windows overlap in time, so information leaks across a train/test boundary unless observations near it are purged or embargoed (López de Prado, 2018). For dependent observations, resampling individual points underestimates variance; block bootstrap methods resample contiguous segments instead (Künsch, 1989). Our corrected protocol applies both ideas.

3 Data and Target

3.1 Data

We use daily Open-High-Low-Close prices and volumes from Yahoo Finance (`yfinance`, dividend-adjusted) for SPY, DIA, QQQ, GLD and TLT, together with the CBOE Volatility Index (VIX), from 3 January 2000 to 30 December 2024. GLD and TLT start later (18 November 2004 and 30 July 2002). The models are trained on SPY only. Table 1 summarizes the samples after removing the first rows, for which the 200-day average and other rolling windows are undefined.

Adjusted prices returned by `yfinance` differ slightly between successive downloads: on SPY, 5,378 of 6,288 closing prices differed between two calls, by at most \$0.0002. All results below use a single cached snapshot of the data so that runs are reproducible. Section 5.7 measures how much these differences matter.

Asset	Train rows	Test rows	Test period	Positive rate (train / test)	Crossovers in test
SPY	4,563	1,521	2018-12-13 – 2024-12-30	11.6% / 9.9%	5
DIA	4,563	1,521	2018-12-13 – 2024-12-30	15.1% / 10.3%	6
QQQ	4,563	1,521	2018-12-13 – 2024-12-30	14.4% / 8.7%	5
GLD	3,643	1,215	2020-03-04 – 2024-12-30	13.1% / 22.5%	10
TLT	4,080	1,360	2019-08-06 – 2024-12-30	20.3% / 19.4%	9

Table 1: Samples after a chronological 75/25 split per asset. Positive rate is the share of days labeled 1. Crossovers are counted as events, not rows.

3.2 A first target based on future returns

We first labeled day t as bullish when the return over the next h days exceeded a threshold r:

$$y_t = \mathbf{1}\left[\frac{P_{t+h} - P_t}{P_t} > r\right], \tag{1}$$

over the grid $h \in \{15, 30, 60, 90\}$ days and $r \in \{3\%, 5\%, 7\%, 10\%\}$. The configuration $h = 60$, $r = 5\%$ labels 36.7% of days as bullish but switches 440 times between 2000 and 2024, far more than the number of genuine market regimes. It also misses sustained trends: only 33.8% of days in the 2013–2017 bull market are labeled bullish (Figure 4). We therefore abandoned this target.

3.3 Moving-average regime and classification target

The regime indicator is

$$s_t = \mathbf{1}\left[\mathrm{MA}_{50}(t) > \mathrm{MA}_{200}(t)\right], \tag{2}$$

where $\mathrm{MA}_k(t)$ is the k-day simple moving average of the closing price. On SPY it changes 24 times over the period where both averages are defined, labels 71.9% of days as bullish, and covers 89.5% of days in 2013–2017 (Figure 5).

The classification target flags the 30 days that precede a regime change:

$$y_t = \mathbf{1}\left[\exists\, i \in \{1, \ldots, 30\} : s_{t+i} \neq s_t\right]. \tag{3}$$

Positive days form contiguous blocks of 30, one block per crossover (two crossovers less than 30 days apart produce a merged block). Over the whole SPY sample, 11.1% of days are positive. The last 30 days of the sample cannot be labeled positive, since crossovers after 30 December 2024 are not observed.

Appendix A checks that the regime label s_t is stable under multiplicative price noise of the size of daily price moves.

4 Methodology

4.1 Features

We use 12 features computed from past data only (Table 2). Raw price levels are excluded because they are period-specific. Three features are expressed in price units: ATR, MA velocity, and MA_cross_momentum, whose numerator is a difference of MA_{50} values in dollars. This matters when a model trained on SPY is applied to assets with different price levels.

The moving-average features are

$$g_t = \frac{\mathrm{MA}_{50}(t) - \mathrm{MA}_{200}(t)}{\mathrm{MA}_{200}(t)}, \tag{4}$$

$$\mathrm{MA_velocity}_t = \left[\mathrm{MA}_{50}(t) - \mathrm{MA}_{50}(t-5)\right] - \left[\mathrm{MA}_{200}(t) - \mathrm{MA}_{200}(t-5)\right], \tag{5}$$

$$\mathrm{MA50_slope}_t = \frac{\mathrm{MA}_{50}(t) - \mathrm{MA}_{50}(t-10)}{\mathrm{MA}_{50}(t)}, \tag{6}$$

$$\mathrm{Distance_normalized}_t = \frac{g_t}{\frac{1}{50}\sum_{i=0}^{49} \sigma_{t-i}}, \tag{7}$$

$$\mathrm{MA_cross_momentum}_t = \frac{\Delta_5 \mathrm{MA}_{50}(t) - \Delta_{10}\mathrm{MA}_{50}(t)}{|g_t|}, \tag{8}$$

where $\Delta_k \mathrm{MA}_{50}(t) = \mathrm{MA}_{50}(t) - \mathrm{MA}_{50}(t-k)$ and σ_t is the 20-day realized volatility. The sign of g_t equals the current regime s_t; this is not a leak of the target, which concerns future changes of s_t, but it means the models know the current regime.

Feature	Definition
Return	1-day return
Volatility	standard deviation of daily returns over 20 days
Cumulated_Return_5d	compounded return over 5 days
RSI14	relative strength index, 14 days, simple averages
Volume_ROC	14-day percentage change in volume
ATR	average true range, 14-day simple average
VIX_spike	VIX_t divided by its 60-day moving average (including t)
Distance_GC	g_t
MA_velocity, MA50_slope, Distance_normalized, MA_cross_momentum	see equations above

Table 2: Feature set.

4.2 Models and hyperparameters

We train an XGBoost classifier (Chen and Guestrin, 2016) with `scale_pos_weight` equal to the ratio of negative to positive training days, and a Random Forest with balanced class weights. Hyperparameters were selected by randomized search (100 configurations, `TimeSeriesSplit` with 5 folds, F1 on the positive class) on the SPY training set only. For XGBoost, early stopping with a patience of 30 rounds on the last validation fold reduced the number of trees from 327 to 138. Because this validation fold is also scored by the search, the selected XGBoost configuration is slightly optimistic. Tables 3 and 4 report the values used in all experiments; the search is in notebook `02` of the repository.

Hyperparameter	Value
n_estimators	138
max_depth	5
learning_rate	0.043
subsample	0.727
colsample_bytree	0.698
gamma	0.054
min_child_weight	1

Table 3: XGBoost (CV F1 0.659).

Hyperparameter	Value
n_estimators	346
max_depth	9
min_samples_split	17
min_samples_leaf	5
max_features	0.546

Table 4: Random Forest (CV F1 0.638).

4.3 Evaluation protocols

Each asset is split chronologically: the first 75% of rows form the training set, the last 25% the test set. Models are trained on SPY and applied unchanged to the other assets, using the threshold selected on SPY. We compare two protocols.

Naive protocol. The model is trained on the full SPY training set. The decision threshold is the value in $\{0.10, 0.15, \ldots, 0.95\}$ that maximizes F1 on the SPY test set. Confidence intervals come from a bootstrap that resamples individual days. This reproduces the first version of this study.

Corrected protocol. The last 20% of the SPY training set (913 days, 30 April 2015 to 12 December 2018, containing 5 crossovers) is held out for validation. A model trained on the earlier days, minus a 30-day embargo before the validation block, selects the threshold that

maximizes validation F1. The final model is then trained on the full training set minus its last 30 days, whose labels depend on test-period prices (López de Prado, 2018). Confidence intervals come from a moving-block bootstrap with blocks of 30 days (Künsch, 1989). Both bootstraps use $B = 1000$ resamples and percentile intervals at 95%.

Intermediate protocols. The two protocols differ in two respects: where the threshold is selected, and whether the 30 days before each boundary are dropped. To separate them, we evaluate two intermediate variants. In variant (a), the threshold is selected on the SPY test set, as in the naive protocol, but the model is trained without the last 30 training days, as in the corrected protocol. In variant (b), the threshold is selected on the validation block, as in the corrected protocol, but without any embargo: the inner model is trained on all days before the validation block and the final model on the full training set.

Paired comparison. To compare the extrapolation rule with a model on a given asset, we resample the same 30-day blocks for both methods, compute the difference of their F1 scores on each resample ($B = 2000$), and report the 95% percentile interval of that difference.

Event-level metrics. F1 on daily labels counts each crossover 30 times and does not show when alerts start. For each crossover, the lead time is the number of days between the first alert in the 30 days before it and the crossover; by construction it cannot exceed 30. A false-alarm day is an alert on a negative day. A false-alarm episode is a run of consecutive alert days that contains no positive day.

4.4 Baselines

Extrapolation rule. Let $\dot{g}_t = (g_t - g_{t-5})/5$. If the gap kept its recent trend, it would reach zero after $\tau_t = -g_t/\dot{g}_t$ days. The rule predicts

$$\hat{y}_t = \mathbf{1}\left[0 < \tau_t \leq 30\right]. \tag{9}$$

The lookback of 5 days was fixed by hand and not tuned. The horizon is set equal to the target horizon; as a check, a search over $\{5, 10, \ldots, 90\}$ on the SPY training set also selects 30. No parameter of the rule is fitted to data. Its rationale is mechanical: $\mathrm{MA}_{200}(t + 30)$ shares 170 of its 200 prices with information available at t.

For threshold-free comparisons we use the continuous score $1/(1 + \tau_t)$, set to 0 when no crossover is predicted ($\tau_t \leq 0$ or $\dot{g}_t = 0$). It orders days in the same way as $-\tau_t$, and the binary rule corresponds to a score of at least $1/31$.

Stratified random. Predicting 1 at random with the training positive rate p gives an expected F1 of $2pq/(p + q)$, where q is the test positive rate.

4.5 Additional analyses

Walk-forward. On the full SPY series, we use an expanding training window starting at 50% of the sample and five test folds of 608 days. In each fold, the threshold is re-selected on the last 20% of the current training window with the same embargo. Folds with fewer than two positive days are skipped.

Numerical sensitivity. We multiply all SPY, DIA and QQQ prices by $(1+\varepsilon_t)$, $\varepsilon_t \sim \mathcal{N}(0, 10^{-12})$, i.e. relative perturbations of order 10^{-6}, similar to the differences between two `yfinance` downloads, and rerun the corrected protocol 20 times.

5 Results

5.1 Naive protocol

Under the naive protocol, the test-optimal thresholds are 0.90 for XGBoost and 0.65 for Random Forest. Mean F1 over SPY, DIA and QQQ is 0.738 for XGBoost and 0.720 for Random Forest (Table 5). These figures are close to those of the first version of this study, which used another download of the same data and found a test-optimal XGBoost threshold of 0.85.

Asset	XGBoost ($\tau = 0.90$)	Random Forest ($\tau = 0.65$)
SPY	0.762 [0.698, 0.815]	0.763 [0.709, 0.810]
DIA	0.683 [0.618, 0.740]	0.653 [0.592, 0.706]
QQQ	0.769 [0.705, 0.824]	0.745 [0.685, 0.798]
GLD	0.661 [0.600, 0.713]	0.737 [0.693, 0.776]
TLT	0.618 [0.558, 0.676]	0.643 [0.596, 0.683]
Mean SPY/DIA/QQQ	0.738	0.720

Table 5: Naive protocol: F1 with observation-level bootstrap 95% intervals. Thresholds are selected on the SPY test set.

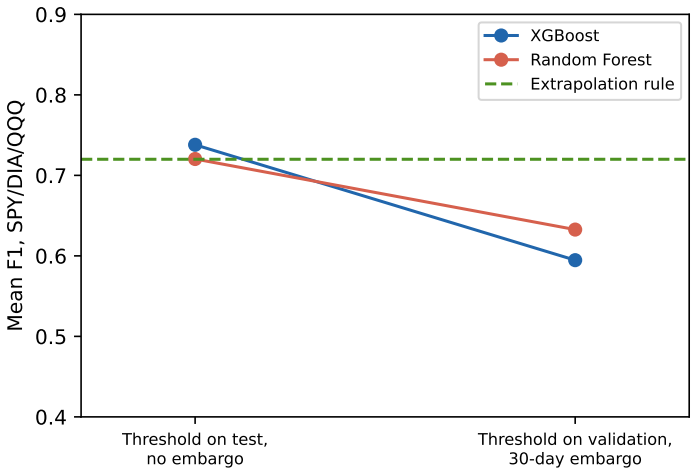


Figure 1: Mean F1 over SPY, DIA and QQQ under the naive and corrected protocols, compared with the extrapolation rule.

5.2 Corrected protocol

On the validation block, the selected thresholds are 0.20 for XGBoost (validation F1 0.792) and 0.45 for Random Forest (0.760). Table 6 reports test results. Mean F1 over SPY, DIA and QQQ falls to 0.595 for XGBoost and 0.633 for Random Forest (Figure 1). The drop comes from precision: with the lower thresholds, XGBoost flags 291 SPY test days for 150 positive days, against 139 for the extrapolation rule.

Table 7 isolates the two changes between the protocols. Moving threshold selection from the test set to the validation block lowers mean F1 by 0.077 to 0.103 for Random Forest and by 0.125 to 0.157 for XGBoost, depending on whether the embargo is applied. The embargo changes mean F1 by -0.010 or $+0.016$ for Random Forest and by -0.018 or $+0.014$ for XGBoost. Under price perturbations of one part in a million, the XGBoost mean F1 varies by up to 0.047 (Section 5.7),

Asset	Model	Precision	Recall	F1 [95% block CI]	Days flagged
SPY	XGBoost	0.457	0.887	0.603 [0.384, 0.771]	291
	Random Forest	0.535	0.820	0.647 [0.426, 0.794]	230
	Extrapolation	0.748	0.693	0.720 [0.489, 0.865]	139
	Stratified			0.106	
DIA	XGBoost	0.415	0.834	0.554 [0.324, 0.692]	316
	Random Forest	0.533	0.828	0.648 [0.409, 0.782]	244
	Extrapolation	0.609	0.586	0.597 [0.382, 0.769]	151
	Stratified			0.109	
QQQ	XGBoost	0.468	0.947	0.627 [0.393, 0.788]	267
	Random Forest	0.457	0.886	0.603 [0.341, 0.766]	256
	Extrapolation	0.853	0.833	0.843 [0.678, 0.931]	129
	Stratified			0.099	
GLD	XGBoost	0.643	0.971	0.774 [0.643, 0.877]	412
	Random Forest	0.696	0.938	0.799 [0.679, 0.891]	368
	Extrapolation	0.786	0.740	0.762 [0.637, 0.860]	257
	Stratified			0.153	
TLT	XGBoost	0.459	0.811	0.586 [0.413, 0.711]	466
	Random Forest	0.526	0.780	0.628 [0.448, 0.748]	392
	Extrapolation	0.783	0.723	0.752 [0.583, 0.853]	244
	Stratified			0.145	
Mean SPY/DIA/QQQ				XGB 0.595 / RF 0.633 / Extr. 0.720	

Table 6: Corrected protocol: test results with 30-day moving-block bootstrap intervals. Stratified is the expected F1 of random predictions.

so the XGBoost differences in this table are measured with that much noise. Random Forest varies by 0.002 under the same perturbations and gives the more reliable measure: almost all of the protocol effect comes from the source of the threshold.

		XGBoost		Random Forest	
Threshold selected on	Embargo	τ	F1	τ	F1
Test set (naive)	no	0.90	0.738	0.65	0.720
Test set, variant (a)	yes	0.85	0.720	0.70	0.710
Validation, variant (b)	no	0.25	0.581	0.45	0.617
Validation (corrected)	yes	0.20	0.595	0.45	0.633

Table 7: Mean F1 over SPY, DIA and QQQ when the two elements of the protocol are changed one at a time. τ: selected threshold.

5.3 Comparison with the extrapolation rule

The extrapolation rule obtains the highest F1 on SPY, QQQ and TLT. Random Forest is ahead on DIA, and both models are ahead on GLD (Figure 2). All methods are far above the stratified baseline. The block-bootstrap intervals of the three methods overlap on every asset. Overlapping marginal intervals do not show that two methods perform alike: both are scored on the same crossovers, so their F1 values move together across resamples, and the interval of their difference can exclude zero while the marginal intervals overlap.

Table 8 reports the paired comparison. The rule is better than both models on QQQ (+0.216 $[0.053, 0.422]$ against XGBoost, +0.240 $[0.073, 0.452]$ against Random Forest) and better than

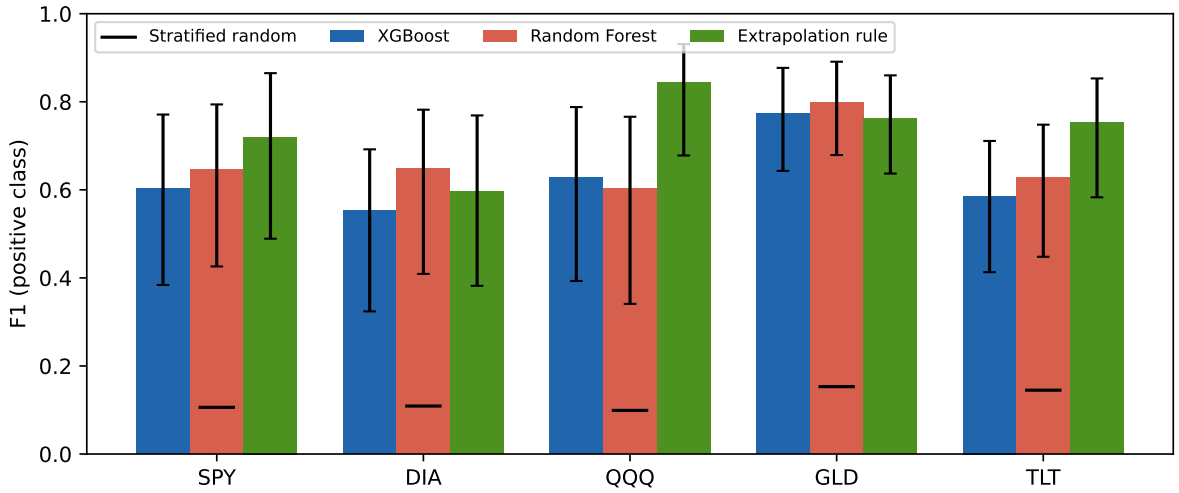


Figure 2: Test F1 per asset under the corrected protocol, with 30-day block bootstrap 95% intervals. Black marks: expected F1 of stratified random predictions.

XGBoost on TLT (+0.166 [0.003, 0.341]). The seven other intervals contain zero, including the three where a model has the higher F1. No comparison favors a model. These are ten tests without correction for multiplicity, and the TLT lower bound is close to zero.

Asset	Rule − XGBoost	Rule − Random Forest
SPY	+0.117 [−0.071, 0.312]	+0.072 [−0.115, 0.262]
DIA	+0.043 [−0.111, 0.241]	−0.051 [−0.185, 0.132]
QQQ	+0.216 [0.053, 0.422]	+0.240 [0.073, 0.452]
GLD	−0.011 [−0.144, 0.111]	−0.036 [−0.168, 0.087]
TLT	+0.166 [0.003, 0.341]	+0.124 [−0.049, 0.310]

Table 8: Difference in test F1 between the extrapolation rule and each model, with 95% intervals from a paired 30-day block bootstrap ($B = 2000$, same resampled days for both methods).

Threshold-free ranking quality on SPY, measured by average precision, is 0.790 for XGBoost, 0.726 for Random Forest and 0.827 for the continuous score of the rule, against a prevalence of 0.099 (Figure 6). The rule's score has the highest average precision on SPY, QQQ (0.893) and TLT (0.797), the lowest on DIA (0.602, against 0.697 and 0.663), and lies between the two models on GLD (0.818, against 0.827 and 0.802). The model scores carry information, but on most assets they do not rank days better than the rule.

5.4 Event-level metrics

Every method raises at least one alert within the 30 days before each of the 35 test crossovers. Table 9 reports when the alerts start and how many are false. On SPY, the first alert comes a median of 30 days before the crossover for both models and 17 days for the rule, and the rule produces 35 false-alarm days against 158 for XGBoost and 107 for Random Forest. A median lead of 30 days is the ceiling set by the 30-day window: the model was already alerting when the window opened. The models alert earlier than the rule on SPY and DIA. On QQQ and GLD all three methods reach the ceiling, and on TLT the rule's median lead (27 days) is one day longer than the models'.

The models pay for their early alerts with false alarms. Pooled over the five assets, XGBoost and Random Forest produce 884 and 658 false-alarm days against 221 for the rule, 4.0 and 3.0

times as many; per asset the ratio ranges from 1.9 to 7.5. Counted as episodes, the totals are 63, 36 and 20.

Asset	Crossovers	Median lead (days)	False-alarm days	False-alarm episodes / year
SPY	5	30 / 30 / 17	158 / 107 / 35	1.82 / 1.33 / 0.99
DIA	6	24.5 / 24 / 15.5	185 / 114 / 59	3.15 / 1.99 / 0.50
QQQ	5	30 / 30 / 30	142 / 139 / 19	1.82 / 0.99 / 0.33
GLD	10	30 / 30 / 30	147 / 112 / 55	1.24 / 0.41 / 0.83
TLT	9	26 / 26 / 27	252 / 186 / 53	2.96 / 1.48 / 0.93

Table 9: Event-level metrics on the test sets, in the order XGBoost / Random Forest / extrapolation rule. Lead times are capped at 30 days by the alert window. Years are counted as 252 trading days.

5.5 Uncertainty

Block intervals are 2.6 to 3.7 times wider than observation-level intervals (3.3 on average over the 15 asset-method pairs). For XGBoost on SPY, the observation-level interval is $[0.547, 0.654]$ and the block interval $[0.384, 0.771]$. Positive days come in blocks of 30, so resampling individual days treats one crossover as 30 independent observations. With 5 to 10 crossovers per test set, one missed or spurious event changes F1 substantially.

5.6 Walk-forward validation

Table 10 reports the walk-forward results on SPY. Fold 1 (November 2012 to April 2015) contains no positive day and is skipped. Pooled F1 over the four remaining folds is 0.651 for XGBoost and 0.638 for Random Forest (Figure 7). Fold F1 ranges from 0.387 to 0.792 and the selected threshold from 0.10 to 0.80. Fold 5 contains a single crossover.

Fold	Train size	Positive days	Crossovers	XGBoost τ / F1	Random Forest τ / F1
2 (2015-04 – 2017-09)	3,650	115	4	0.10 / 0.792	0.10 / 0.691
3 (2017-09 – 2020-02)	4,258	68	2	0.15 / 0.562	0.40 / 0.642
4 (2020-02 – 2022-07)	4,866	82	3	0.80 / 0.790	0.75 / 0.725
5 (2022-07 – 2024-12)	5,474	30	1	0.40 / 0.387	0.45 / 0.420
Pooled				0.651	0.638

Table 10: Walk-forward validation on SPY, expanding window, threshold re-selected in each fold.

5.7 Numerical sensitivity

Over 20 runs with relative price perturbations of order 10^{-6}, mean F1 over SPY, DIA and QQQ is 0.609 ± 0.012 for XGBoost (range 0.596–0.643), 0.634 ± 0.001 for Random Forest and 0.720 for the extrapolation rule, which does not change. The selected XGBoost threshold varies between 0.20 and 0.35. The likely source is `MA_cross_momentum`, which divides by $|g_t|$ and amplifies small differences near a crossover.

6 Discussion and Limitations

What the results support. Crossovers of the 50- and 200-day averages can be flagged 30 days in advance far better than chance. Most of this predictability follows from the construction of the averages: a linear extrapolation of their gap reaches an F1 of 0.720 on SPY, DIA and QQQ, and neither classifier exceeds it once the threshold is selected without access to the test set. The paired test finds no asset where a model is significantly better than the rule. The models start alerting earlier on two of the five assets, at the cost of three to four times as many false-alarm days overall.

Protocol effects. Moving from the naive to the corrected protocol lowered mean F1 by 0.143 for XGBoost and 0.087 for Random Forest. Most of this comes from selecting the threshold on the test set; the embargo has an effect below 0.02 in either direction. Observation-level bootstrap intervals were about three times too narrow. Both problems are easy to miss when labels overlap in time.

Limitations.

- Each test set contains 5 to 10 crossovers, and the SPY, DIA and QQQ test sets cover the same dates on highly correlated assets. They are not independent confirmations.
- The selected threshold depends strongly on the period used for validation, as shown by the walk-forward folds.
- XGBoost results are sensitive to negligible numerical perturbations of the input data.
- Hyperparameters come from a search whose early stopping used a fold that the search also scored.
- Hyperparameters were selected on the SPY training set, which ends in December 2018. It overlaps walk-forward folds 2 and 3, so these two folds are not fully out of sample for the hyperparameters.
- ATR, MA velocity and MA_cross_momentum are expressed in price units (the numerator of MA_cross_momentum is in dollars), which affects transfer to other assets.
- Lead times are capped at 30 days by the alert window, so they cannot show whether a model anticipates a crossover by more than 30 days.
- The study evaluates classification only. It does not evaluate a trading strategy, transaction costs or returns.

Future work. A meaningful test of machine learning on this task should start from the extrapolation rule and ask whether additional information, such as features unrelated to the moving averages themselves, reduces false alarms. Longer horizons, where the mechanical overlap between present and future averages is smaller, would also make the task less trivial.

7 Conclusion

A first evaluation suggested that XGBoost predicts Golden and Death Crosses 30 days ahead with an F1 of about 0.74 and transfers across ETFs. With the threshold selected on validation data, an embargo at the train/test boundary and block bootstrap intervals, F1 falls to about 0.6, and an extrapolation of the moving-average gap with no fitted parameter reaches 0.72, with three to four times fewer false-alarm days than the models. The modest result that remains

is that crossover timing at this horizon is largely mechanical, and that evaluation choices can easily create the impression of a learned signal.

Reproducibility. Code and all result tables are available at `https://github.com/Bissapinho/Trend-classifier`. `scripts/run_main.py` produces every number of Section 5 except Section 5.7, which comes from `scripts/perturbation.py`; Section 3 and Appendix A come from `scripts/target_analysis.py`.

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A Robustness of the regime label

We multiplied each SPY closing price by $(1 + \varepsilon_t)$, with $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ independent across days, recomputed s_t, and repeated this 50 times for each of 50 values of $\sigma \in [0.004, 0.20]$. The noise is proportional to each day's price and uses no information from other dates. For reference, the standard deviation of SPY daily returns is 0.0122. At σ equal to that value, 0.4% of labels change and the share of bullish days moves by 0.09 percentage points. Over the whole grid, the share of labels that change rises from 0.1% to 8.4% and the share of bullish days moves by at most 1.9 percentage points (Figure 3). The regime label is therefore stable under noise of the size of daily price moves.

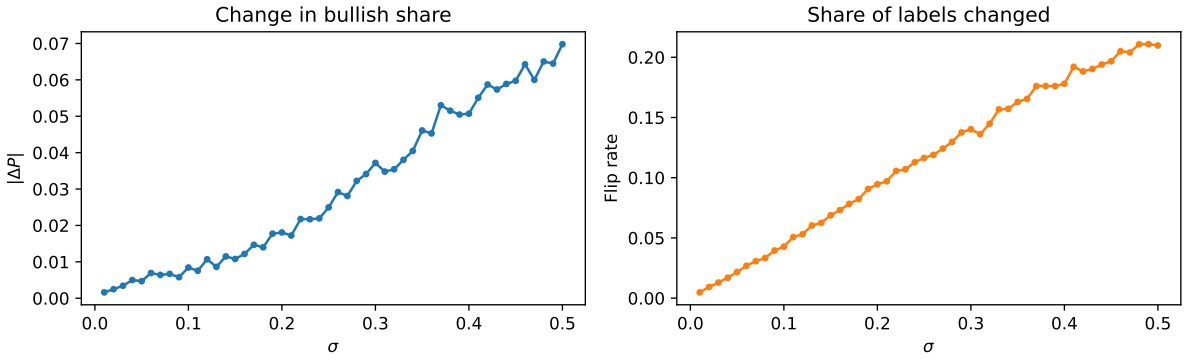


Figure 3: Robustness of the regime label to multiplicative price noise: change in the share of bullish days (left) and share of labels that change (right), averaged over 50 draws per σ.

B Additional figures

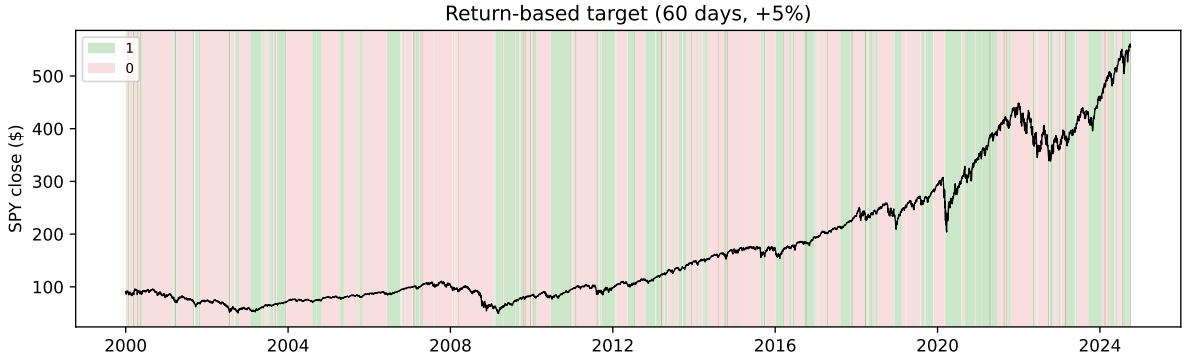


Figure 4: Return-based target on SPY ($h = 60$ days, $r = 5\%$): 440 label switches over 2000–2024.

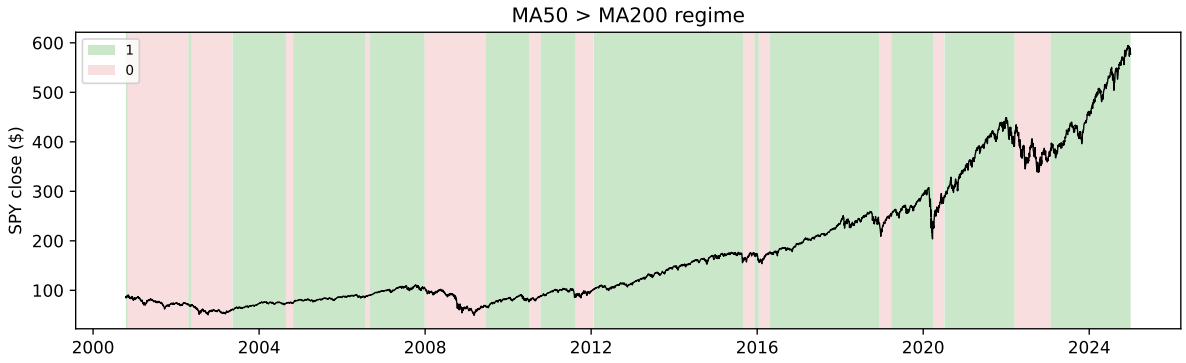


Figure 5: MA50/MA200 regime on SPY: 24 crossovers.

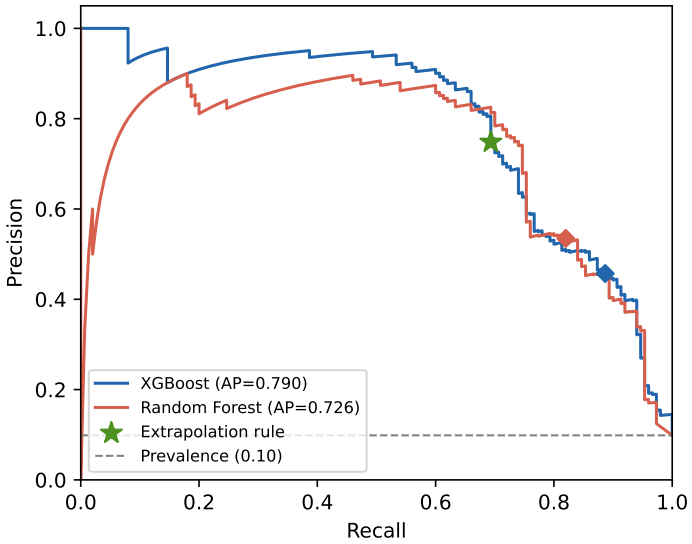


Figure 6: Precision-recall curves on the SPY test set. The extrapolation rule's curve uses the continuous score $1/(1+\tau_t)$. Diamonds: operating points at the validation-selected thresholds. Star: binary extrapolation rule ($\tau_t \leq 30$). AP: average precision.

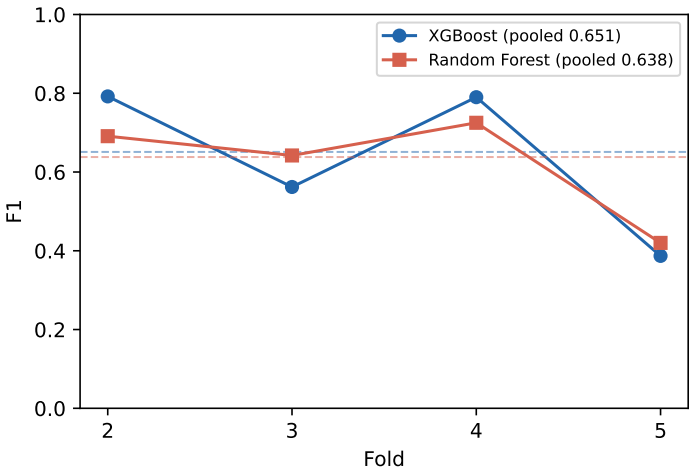


Figure 7: Walk-forward F1 per fold on SPY. Dashed lines: pooled F1.